DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

2. TITLE	1. IDENTIFICATION NUMBER
COMMERCIAL MANUALS FOR INDUSTRIAL PLANT EQUIPMENT NUMERIC CONTROL	DI-TMSS-80384

3. DESCRIPTION/PURPOSE

3.1 These commercial manuals are operating and shop maintenance instructions which enable a journeyman mechanic to maintain, service, and repair a Numeric Control (NC) and interface assembly installed on Industrial Plant Equipment (IPE). These manuals shall include the following: (a) repair part listings that provide positive identification of replaceable parts and components required for the repair of the NC and interface assembly; (b) detailed wiring diagrams and schematics of the electrical circuitry of the NC and interface assembly.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC REQUIRED	6b. GIDEP REQUIRED
870715	S/DIPEC-SSA		

7 APPLICATION/INTERRELATIONSHIP

7.1 This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

7.2 This DID will be used when numerically controlled IPE is retrofitted with a new NC. This process normally requires reverse engineering to interface the NC with the item of IPE. Also manuals, drawings, and schematics applicable to the new NC are required for the operation, maintenance, set-up, and programming. This DID is used in the IPE maintenance program.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		S4149

10. PREPARATION INSTRUCTIONS

10.1 Format.

10.1.1 Manuals may be in separate books or volumes consistent with the contractor's standard commercial practice. Manufacturer's manuals or data for components, assemblies, etc., which are provided as part of this contract, shall be assembled and bound securely into a manageable volume or volumes and shall be indexed by major assembly and component in sequential order. Where practicable, all drawings and instruction sheets should not exceed a single foldout sheet.

10.1.2 A table of contents (index) shall be made part of the assembled publication set.

10.1.3 Pen and ink changes or rough penciled diagrams or illustrations normally furnished with the preliminary draft shall not be included in the finished product.

10.1.4 The cover sheet or binder on each volume of the manuals shall be identified and marked as follows:

a. Numerical control manufacturer and address.
b. Contractor's name and address.
c. Contract number.
d. Equipment identification, make, model, and serial number of the numeric control.
e. Make, model, and serial number of the equipment on which the numeric control is installed.
f. Volume number and title of manual.

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11 DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Block 10, Preparation Instructions (continued)

10.2 Contents of the Commercial Manuals:

a. Operating instructions.

b. Programming instructions and programs.

c. Maintenance, service, and repair instructions.

d. Parts List.

e. Installation and assembly instructions.

10.2.1 Operating instructions. The operating instructions shall include the following:

a. Operator prestart checks.
b. Starting procedure and controls.
c. Operation of the basic NC.
d. Safety precautions to be observed while operating under all operating conditions for which the NC was designed.

10.2.2 Programming instructions and programs. The programming instructions shall include specific instructions and illustrations of the programming function of the specific controls as follows:

a. Machine application.
b. Operational and functional description.
c. Programming language (if applicable).
d. Programming and sample program(s).
e. Control features.
f. Operating procedures.
g. Modes of operation.
h. Procedural summaries, diagnostics, error messages, and recovery procedures (as applicable).
i. Tape preparation and application (if used).
j. Executive programs in punched tape, magnetic tape or disk, hard copy, and laser media (as applicable).

10.2.3 Maintenance, service, and repair instructions. The contents of the maintenance, service, and repair instructions shall include all necessary instructions that will enable an average skilled mechanic, without prior knowledge of the specific type, make, or model of equipment, to maintain the equipment in a safe and serviceable condition. These instructions shall contain all necessary illustrations, charts, and diagrams covering the following:

a. A table of preventive maintenance instructions, including frequency in time or hours covering routine servicing and adjustments.
b. Troubleshooting guides and diagnostic techniques that will enable prompt isolation of the cause of malfunction with corrective maintenance instructions.

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Block 10, Preparation Instructions (continued)

c. Instructions for removal, replacement, disassembly, and assembly of all components, assemblies, and subassemblies, accessories, and attachments normally subjected to wear, damage, or malfunctioning.

d. Maintenance and repair data, i.e., tolerances, dimensions, settings, and adjustments, normally required for performing routine maintenance servicing, including removal, replacement, and reassembly of components, assemblies, subassemblies, accessories, and attachments normally subject to wear, damage or malfunctioning.

10.2.4 Parts List. The parts list shall provide positive identification and coverage of all of the parts of components, assemblies, subassemblies, and accessories of the end item normally subjected to wear, malfunctioning, damage, or loss. Common hardware items such as mild steel bolts, nuts, washers, cotter pins, resistors, capacitors, relays, integrated circuits and the like may be omitted.

a. Clear and legible illustrations, drawings, and exploded views shall be provided to identify all individual parts, components, assemblies and subassemblies of the end item. Part numbers and description shall be shown on illustrations or listed separately. When the illustrations omit the part numbers and description, both the illustrations and separate listing shall show the index, reference, or key number which will cross-reference the illustrated part or listed part. Parts shown in the listings shall be grouped by components, assemblies, and subassemblies with individual parts identified to the assembly.

b. The parts list shall include the prime contractor's part number for purchased production parts, if such parts are altered to meet the prime contractor's design configuration. (Repainting, marking, or other nonsignificant operations are not adequate cause for use of exclusively assigned numbers.)

c. Components, assemblies/parts, purchased by the prime contractor for which the prime contractor does not have design control, shall be identified by the actual manufacturer's name and part number. Detail parts in a prime contractor's assembly, as well as attaching parts, for which the prime contractor does not have design control shall also be identified by the applicable actual manufacturer's part numbers. This paragraph does not restrict prime contractors from assigning their part numbers as long as the actual manufacturer's part number and manufacturer's name are shown.

d. Contractor may elect to append an annex of cross-reference numbers to implement 10.2.4.c. (above) requirements when implementation in manual form varies drastically with the style, format, and method of contractor's standard commercial practice.

Subject cross-reference in an annex will appear in the following format:

Contractor's Part No. Alpha Numeric Seq.	Actual Manufacturer Name	Actual Manufacturer Part No.
1000001	John Doe & Co.	2000002

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Block 10, Preparation Instructions (continued)

e. Parts lists may cover more than one model or series of numeric controls, components, assemblies, subassemblies, attachments or accessories, such as master parts catalog, in accordance with the contractor's standard commercial practice or option. Deletion of part numbers not applicable to equipment under the contract is not necessary, provided those parts applicable to equipments being furnished are clearly identified in the manual provided. Identification of the part shall be such that all parts may be ordered and centrally stocked by the government without further identification to the make, model, and serial number of the equipment being provided under the contract.

10.2.5 Installation and assembly instructions. The installation and assembly instructions shall be comprehensive enough to enable a journeyman mechanic to install a specific numeric control on an item of IPE. They shall include parts lists, electrical drawings, mechanical drawings, assembly drawings, and interface drawings which will depict the final configuration and wiring of the numeric control to the item of IPE.

10.3 Approval and changes. Change sheets or revised manuals shall be required if any components, assembly, subassembly, attachment, or accessory of the equipment is changed subsequent to delivery and acceptance of the approved commercial manual. These change sheets or revised manuals are required after determination of changes to components, etc.

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